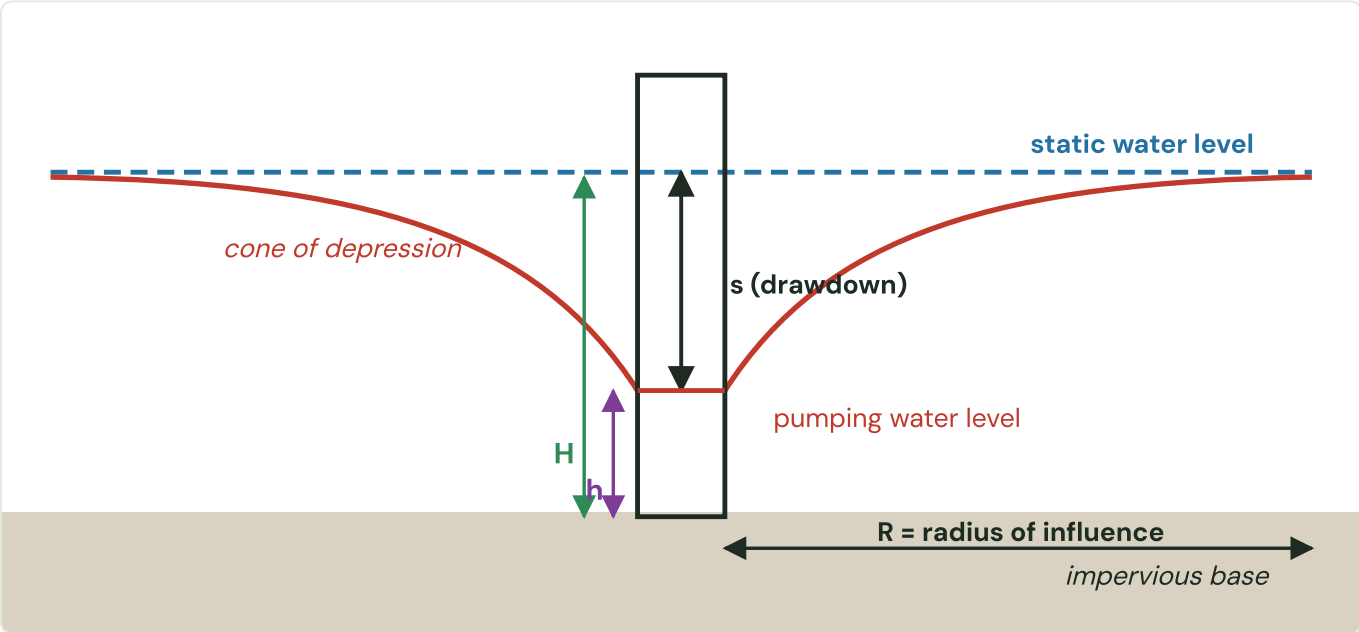
WATER RESOURCES, WELLS & PUMPS · NOTE WL02

Well Hydraulics and Tube-Well Design — Drawdown, Yield Equations, Screens and Gravel Pack

Course topics: IDE6.2 L10, IDE6.2 L11, IDE6.2 L13, IDE-510 T5, IDE-510 T7, IDE-510 T10, IDE-506 T8

Terms

Term	Meaning
Static water level	Water level in the well before pumping — the water table (unconfined) or the piezometric surface (confined)
Pumping water level	The stabilised level in the well during pumping
Drawdown	Vertical distance the level is lowered by pumping
Cone of depression	The inverted cone the water table or piezometric surface takes around a pumped well
Radius of influence (R)	Horizontal distance from the well centre within which the original surface is lowered



Pumped well in an unconfined aquifer: drawdown, cone of depression and radius of influence

Soil formation and texture (Table 5.1, after Bennison, 1947)	Radius of influence (m)
Fine sand with some silt and clay	30 – 90
Fine to medium sand, fairly clean	90 – 180
Coarse sand and fine gravel	180 – 300
Coarse sand and gravel without silt and clay	300 – 600

Flow through an aquifer depends on the slope of the water table (or piezometric surface) and on the **permeability** — the ease with which the material transmits water, expressed as **hydraulic conductivity** in cm per hour or m per day.

Yield of a well (steady flow, fully penetrating)

$$\text{Unconfined aquifer: } Q = K\,\pi\,(H^2 - h^2) \,/\, [2.3 \log (R/r)]$$

$$\text{Confined aquifer of thickness T: } Q = 2\,\pi\,K\,T\,(H - h) \,/\, [2.3 \log (R/r)]$$

- Q = discharge (m³/day); K = hydraulic conductivity (m/day); H and h = heights of the static and pumping water levels above the aquifer bottom (m); R = radius of influence (m); r = radius of the well (m). 2.3 log converts the natural log (ln = 2.3 $\log_{10}$).
- These are the Dupuit–Thiem equations. Field conditions rarely meet all their assumptions, so treat the result as an approximate yield.

> Unconfined aquifer: K = 20 m/day, H = 30 m, h = 25 m, R = 300 m, r = 0.15 m.
> Q = 20 × π × (900 − 625) / [2.3 × log(2000)] = 17,279 / (2.3 × 3.301) ≈ 2276 m³/day ≈ 26 L/s.

Well characteristics and pump fitting

- For a non-flowing artesian well the drawdown–discharge (H–Q) curve is straight at first; once the aquifer starts to be uncovered it bends upward, because the effective supply area shrinks and the head needed rises quickly.
- In a non-artesian well there is no pressure in the supply zone and the supply area shrinks continuously with pumping, so the H–Q curve rises at an increasing rate.
- **Fitting a pump:** plot the well drawdown curve, superimpose the pump's H–Q curve, and read the discharge at the intersection. Choose the pump that gives the required discharge at maximum efficiency and suits the well. For horizontal centrifugal pumps keep the drawdown within 6 m.

Elements of well design

Performance depends on the aquifer and on design and construction; good design gives optimum performance, long life and reasonable cost. Design covers: depth and spacing of wells; diameter; length and type of screen; effective screen openings; screen location; and the gravel pack. A tube well consists of a pump housing pipe, a pipe assembly to convey water and a screen to admit it.

Element	Design rule
Diameter	Just large enough for the housing pipe, blind pipes and screens; beyond that cost rises disproportionately — doubling the diameter raises yield only ~10% (ICAR eCourse: 60 cm vs 30 cm gives about 12% more, with R = 300 m). Its hydraulic gain: doubling the diameter halves the approach velocity, cuts the drag on particles by ¾ and lowers sand entry
Depth	Take the well to the bottom of the aquifer to use its full thickness — more available drawdown means more yield — unless a lower aquifer has poor-quality water
Screen location and length	Set by aquifer thickness, stratification and available drawdown
Slot size	From the particle-size distribution of aquifer samples taken while drilling
Gravel pack	Not needed in coarse sand and gravel (coir, wire-mesh or continuous-slot screens suffice); provide one in fine formations — it filters fine sand, permits larger slots and higher discharge at safe velocity
Screen diameter	Give enough total slot area to keep the entrance velocity ≈ 3 cm/s (ICAR eCourse: 3–6 cm/s), limiting friction loss, incrustation and corrosion. Entrance velocity = desired yield ÷ total slot area. The screen must resist collapse where cave-in is possible

Design checklist: farm size, cropping pattern and water requirement; aquifer material vs gravel and slot size; discharge–drawdown relation; entrance velocity vs incrustation and corrosion; optimum pipe thickness; well cost, pumping level and yield; pump selection and installation.

Sources: ICAR eCourse 'Groundwater, Wells and Pumps' — Dr M. K. Jha (IIT Kharagpur), reviewed by Dr P. B. S. Sarma (IARI New Delhi). (Lessons 10, 16) Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The distance from a pumped well to where drawdown becomes zero is the

(a) drawdown
(b) radius of influence
(c) cone of depression
(d) specific yield

Q2. Doubling the diameter of a well increases its yield by only about

(a) 2%
(b) 10%
(c) 50%
(d) 100%

Q3. Recommended maximum screen entrance velocity is about

(a) 0.3 cm/s
(b) 3 cm/s
(c) 30 cm/s
(d) 3 m/s

Q4. Radius of influence in coarse sand and gravel without silt and clay is about

(a) 30–90 m
(b) 90–180 m
(c) 180–300 m
(d) 300–600 m

Q5. In the confined-aquifer yield equation, Q is proportional to

(a) $H^2 - h^2$
(b) $H - h$
(c) $(H - h)^2$
(d) H/h

Q6. Doubling a well's diameter reduces drag force on aquifer particles by

(a) ¼
(b) ½
(c) ¾
(d) nothing

Answers

Q1 — (b) radius of influence

Q2 — (b) 10%

Q3 — (b) 3 cm/s

Q4 — (d) 300–600 m

Q5 — (b) H - h

Q6 — (c) ¾ — *Approach velocity halves; drag ∝ v^2.*